# Task 1 Report: Student Project Milestone Tracker

Hypothetical workflow investigation and prototype proposal

Prepared: 24 October 2026
Project type: Hypothetical case study · Synthetic data only

**IMPORTANT:** This report does not claim that the described workflow exists at ICFAI. No real college evidence was collected. Baseline steps, counts, timings and test data are explicitly hypothetical or illustrative.

# 1. Executive summary

This report proposes a Student Project Milestone Tracker for a hypothetical semester-project workflow at ICFAI. The concept centralizes milestone definitions, due dates, submissions, supervisor decisions, revision feedback and an audit trail.

Access to the actual institutional workflow and permission to collect college evidence were unavailable for this exercise. Therefore, the current-state process, sample volume, timings and scenarios in this report are explicitly hypothetical. They must not be interpreted as verified facts about ICFAI or its staff. No actual college observations or measured savings are claimed.

A local HTML/CSS/JavaScript prototype was prepared using synthetic records. It demonstrates milestone status tracking, required-field validation, duplicate detection, demo role checks, revision requests, local audit history and five synthetic test scenarios.

# 2. Scope and method

Scope: a proposed process for student project teams to define milestones, submit deliverables, receive faculty review and revise work when necessary.

Method: desk-based scenario design using the assignment requirements, a clearly labelled hypothetical current-state walkthrough, an illustrative workload model, and synthetic test cases. The report does not claim interviews, campus observations, or access to college systems.

Boundary: the prototype is a demonstration only. It does not integrate with university systems and must not be used with real student records, private messages, marks, signatures or confidential college material.

# 3. Hypothetical current workflow

The following baseline is a scenario for analysis, not a verified description of ICFAI practice:

1. A supervisor communicates project milestones and due dates.

2. Students divide work and record deadlines in their chosen notes or documents.

3. Students submit deliverables through an agreed channel.

4. The supervisor reviews a submission and communicates a decision or feedback.

5. Students revise work when requested and resubmit it.

6. The team tracks remaining work until final submission.

Potential failure modes to test include scattered status information, unclear ownership, missed deadlines, incomplete submissions, repeated submissions and feedback that is difficult to trace. These are hypotheses, not observed campus problems.

# 4. Evidence log and limitations

Because campus access was unavailable, the evidence log records what was actually done in this assignment and separates it from assumptions. It is not an observation log of real college operations.

| Date | Activity / item | Evidence type | Status / limitation |
|---|---|---|---|
| 24 October 2026 | Reviewed the HKAIVERSE Task 1 requirements supplied in the assignment screenshots. | Assignment instructions | Confirmed deliverables and requirement to label a hypothetical walkthrough if access is declined/unavailable. |
| 24 October 2026 | Designed a hypothetical six-step student project milestone workflow. | Hypothetical walkthrough | No campus observation or staff interview; steps are assumptions for design. |
| 24 October 2026 | Created a prototype with synthetic sample records and a five-scenario test runner. | Prototype artifact | Synthetic only; run tests and record actual displayed outcomes before reporting pass/fail. |
| 24 October 2026 | Selected illustrative workload assumptions: 20 teams, 5 milestones/team, 10 minutes current and 5 minutes proposed including review/support per milestone. | Estimate / assumption | Not measured; used only to demonstrate arithmetic and sensitivity. |
| Not available | Actual request volume, real timings, current system and staff workflow. | Unavailable evidence | No access/permission; do not infer that the college lacks a tracker. |

## 5. Proposed workflow

1. Create a project and define each milestone, owner, due date and required deliverable.

2. Validate required fields and prevent duplicate project–milestone–owner combinations.

3. Let a student submit a deliverable reference or description; record the submission and timestamp.

4. Place the milestone in Submitted status for supervisor review.

5. The supervisor approves the submission or requests a revision with a reason.

6. Show current status, deadline, feedback and action history in one dashboard.

7. If saving fails, show an error, preserve the previous record where possible and allow a retry.

Suggested statuses: Not Started, In Progress, Submitted, Revision Required and Approved. Overdue is a derived indicator when the due date has passed and the milestone is not approved.

## 6. Inputs, owners, controls and exceptions

Inputs: project name, milestone title, assigned student, due date, deliverable description/reference and optional description.

Accountable owners: student for preparing and submitting work; faculty supervisor for review and approval; a future system administrator for account and configuration management.

Approval boundary: only an authenticated, authorized supervisor should approve work in a production system. In this prototype, the role selector only simulates that boundary and is not security.

Audit record: event type, milestone ID, actor role, timestamp and a short event description. The browser-based audit log is editable and is not tamper-proof.

Fallback: if a save operation fails, show a clear message and do not claim the action succeeded. A production implementation should use server-side transactions, backups and recoverable error logs.

AI value: AI is not necessary for basic status tracking, required-field checks, duplicate detection or deadline rules. Rules-based software is simpler and more reliable for those tasks. AI might later help summarize non-confidential feedback or flag potentially unclear milestone descriptions, but any suggestions should be reviewed by a human and should not make approval decisions.

## 7. Five synthetic test cases

These cases are included in the prototype's test runner. The expected results below describe the intended behavior. Run the test runner and record the displayed outcome before claiming that a test passed.

| Test | Synthetic input | Expected result | Decision / recovery |
|---|---|---|---|
| 1. Normal request | All required fields present; unique milestone. | Accept and record submission; status becomes Submitted. | Student submits; supervisor reviews. |
| 2. Missing information | Deliverable field empty. | Block submission and identify missing field. | Student corrects the form; no record is created. |
| 3. Duplicate | Same project, milestone title and owner entered twice. | Detect duplicate and preserve the original record. | User edits the existing milestone or changes the genuinely distinct record. |
| 4. Unauthorized approval | Student role attempts approval. | Deny approval and leave status unchanged. | Only supervisor role can review in demo; production needs server-side authorization. |
| 5. Failed step | Empty resubmission or simulated save failure. | Reject empty input; preserve previous submission where possible and show error. | Correct input and retry; production should use transaction/recovery mechanisms. |

## 8. Illustrative workload and ROI calculation

All figures in this section are assumptions for sensitivity analysis, not measured college volumes or timings.

Assumptions: 20 teams per semester; 5 milestones per team; 100 milestone records per semester. Hypothetical current coordination effort is 10 minutes per milestone. Proposed routine coordination effort is 4 minutes per milestone, with an additional 1 minute per milestone for review/support overhead. One-time implementation effort is estimated at 4 hours per semester-project setup, and recurring fixed support at 1 hour per semester.

Current effort = 100 × 10 minutes = 1,000 minutes = 16.67 hours.

Proposed recurring effort = 100 × (4 + 1) minutes = 500 minutes = 8.33 hours.

Recurring net time saved before one-time implementation = 1,000 – 500 = 500 minutes = 8.33 hours per semester.

First-semester net after implementation = 8.33 – 4 – 1 = 3.33 hours saved, under these assumptions.

After the one-time implementation effort has already been paid, estimated net recurring saving is 8.33 – 1 = 7.33 hours per semester.

Illustrative volume range: 10 teams × 5 milestones = 50 records gives 4.17 recurring net hours before implementation/support fixed costs; 30 teams × 5 = 150 records gives 12.5 recurring net hours before fixed costs. The fixed implementation and support assumptions must be subtracted for first-semester net estimates.

These calculations are not proof of actual savings. A real pilot would need permission, measured baseline timing, a defined sample, and comparable before/after measures.

## 9. Risks, unknowns and abandonment condition

Risks: students may not update the tracker; supervisors may prefer existing tools; inaccurate dates may create false overdue flags; local data may be lost or edited; browser role selection is not secure; and a second system could add work rather than reduce it.

Unknowns: whether an equivalent process or system already exists; actual project-team and milestone volumes; current communication channels; current review times; accessibility requirements; and institutional policies for student data.

Evidence that could change the design: a permitted walkthrough showing that an existing official system already handles milestones reliably and users do not experience the hypothesized coordination burden. If this were confirmed, the proposal should shift from building a separate tracker to improving or integrating with the existing approved system—or be abandoned if no material gap remains.

## 10. Prototype description and setup

File: student_milestone_tracker.html. The prototype runs in a modern browser without package installation. Open the file in Chrome or another modern browser.

Choose Faculty Supervisor to create a synthetic milestone. Switch to Student to submit or resubmit. Switch back to Faculty Supervisor to approve or request a revision. Use the filter, audit history and Run 5 test scenarios controls to demonstrate the workflow.

Limitations: local browser storage only; no real authentication, server-side permissions, email notifications, cloud backup, file upload or institutional integration. Do not enter real personal or college data.

## 11. Proposed lab/capstone connection

Host course: to be selected only after checking the student's actual batch curriculum and course prerequisites. A suitable host might be a software engineering, database, web development or capstone course if it exists in the verified curriculum.

Prerequisite knowledge: HTML, CSS, JavaScript, basic data modelling, validation, state/status transitions and testing.

Learning outcomes: model a workflow; define roles and approval boundaries; implement validation and status transitions; test normal and exceptional cases; record limitations; calculate effort using explicit assumptions; and communicate privacy/security risks.

Synthetic dataset: project IDs, milestone titles, fictional student labels, due dates, statuses, submission descriptions and feedback. No real student data is required.

Milestones: (1) requirements and baseline assumptions; (2) workflow and data model; (3) prototype; (4) five test scenarios; (5) evaluation and report.

Acceptance tests: complete request accepted; missing required field blocked; duplicate request detected; student approval attempt denied; failed/empty submission does not overwrite prior data. Students should demonstrate results and explain that

browser-side role checks are not production security.

## 12. Conclusion

The Student Project Milestone Tracker is a feasible prototype for demonstrating workflow design, validation, exception handling and evaluation. Its current-state workflow and ROI are hypothetical, so the report makes no claims about observed ICFAI operations. The next improvement would be to run and record the synthetic tests, refine the interface based on those results, and—only if permission becomes available—validate the assumptions with a faculty member or recent graduate.